

SAINT HUBERT AB, Belgium

WMO: 064760

Lat: 50.0389N

Lon: 5.4039E

Elev: 1930

StdP: 13.70

Time Zone: 1.00 (EUC)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.6	19.2	7.5	8.7	21.5	12.8	11.3	23.4	20.8	37.0	19.0	35.8	8.1	50	0.586

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	80.1	65.4	76.1	63.6	73.0	61.6	66.7	77.0	64.7	74.0	63.0	70.9	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.9	92.3	70.9	61.2	86.6	68.7	59.7	82.2	67.0	32.4	77.2	30.8	74.3	29.4	70.9	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	16.1	14.3	DB	12.0	86.7	5.3	3.7	8.1	89.4	5.0	91.5	2.0	93.6	-1.9	96.3	(4)
(5)			WB	11.3	70.5	5.2	2.0	7.6	71.9	4.5	73.1	1.6	74.2	-2.2	75.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.6	32.1	33.1	39.5	46.3	51.8	57.4	61.3	60.1	54.0	47.7	40.1	34.9		
	DBStd	12.09	6.67	7.75	7.19	7.37	7.06	6.36	6.60	5.86	5.52	6.76	6.35	6.57		
	HDD50	2497	554	473	331	157	63	5	0	1	18	123	302	470		
	HDD65	6800	1019	892	790	560	413	244	156	176	334	535	747	934		
	CDD50	1258	0	1	6	47	118	226	349	316	138	53	4	0		
	CDD65	86	0	0	0	0	3	15	40	25	3	0	0	0		
	CDH74	583	0	0	0	0	15	110	276	169	13	0	0	0		
	CDH80	124	0	0	0	0	0	17	69	38	0	0	0	0		
	WSAvg	7.6	8.4	8.3	8.7	7.4	7.3	7.1	6.5	6.3	6.7	7.5	8.0	8.5		
	PrecAvg	40.9	4.2	3.4	3.1	2.4	2.9	3.1	3.5	3.8	3.0	3.3	3.5	4.6		
(15) Precipitation	PrecMax	50.8	10.6	8.2	6.7	6.3	6.7	6.3	8.3	9.5	6.5	7.7	5.9	10.9		
	PrecMin	30.0	0.5	0.5	0.4	0.2	1.2	0.8	1.7	0.9	0.5	1.4	0.2	0.7		
	PrecStd	5.4	2.0	1.7	1.7	1.5	1.4	1.5	1.5	2.2	1.8	1.3	1.5	2.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.5	55.2	62.3	70.6	75.9	82.4	86.8	84.6	75.8	68.6	57.9	50.3		
		MCWB	47.5	42.0	48.7	55.2	63.2	66.6	66.8	66.5	63.3	57.7	54.6	48.2		
	2%	DB	46.2	48.4	58.2	66.6	71.1	77.1	80.5	78.6	70.4	63.9	53.7	47.8		
		MCWB	45.3	41.1	46.7	52.9	58.3	63.9	65.4	65.0	60.7	57.5	50.8	46.0		
	5%	DB	43.4	44.8	53.6	62.3	67.6	72.9	76.3	74.2	66.8	59.8	51.3	45.8		
		MCWB	42.5	41.7	45.2	51.5	56.8	61.5	63.6	63.2	58.9	55.1	49.2	44.3		
	10%	DB	41.1	42.6	49.3	58.3	63.8	69.3	73.1	70.2	62.9	57.0	48.8	43.6		
		MCWB	40.1	40.4	43.7	49.4	55.3	59.9	61.9	61.0	56.7	53.5	47.2	42.4		
	0.4%	WB	47.7	45.9	50.3	57.1	64.8	69.0	68.8	69.1	64.6	60.1	55.1	49.0		
		MCDB	48.0	49.8	57.4	68.3	74.1	80.3	83.0	79.4	74.1	65.0	57.4	49.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	45.2	43.9	48.2	54.0	61.3	65.3	66.7	65.9	61.8	58.0	51.3	46.6		
		MCDB	46.0	46.1	55.9	63.6	67.3	74.2	77.7	75.3	68.4	62.3	53.1	47.7		
	5%	WB	42.8	42.2	46.2	52.0	58.3	63.0	64.9	63.9	59.7	56.1	49.4	44.6		
		MCDB	43.4	44.2	51.5	60.9	65.1	70.7	74.6	72.4	65.6	59.5	51.1	45.7		
	10%	WB	40.2	40.4	44.5	50.1	55.8	60.9	62.9	62.1	57.7	54.0	47.3	42.5		
		MCDB	40.9	42.6	49.0	57.9	62.5	68.0	71.3	69.0	62.3	56.9	48.9	43.5		
	5% DB	MDBR	6.4	8.0	11.7	14.2	14.6	14.7	14.9	14.1	12.8	9.9	7.1	6.7		
		MCDBR	7.7	12.1	18.4	18.8	19.3	20.0	19.6	19.0	16.9	13.7	10.3	8.4		
		MCWBR	7.4	8.4	9.9	10.2	10.5	10.3	9.3	9.8	9.5	8.6	7.8	7.4		
		MCWBR	7.6	10.2	14.8	17.9	16.5	18.0	17.8	16.6	15.1	12.0	9.2	7.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	7.6	8.0	9.4	10.2	10.1	10.0	9.1	9.7	9.0	8.1	7.5	7.6		
	Clear-Sky Solar Irradiance	taub	0.293	0.316	0.345	0.374	0.383	0.384	0.383	0.377	0.350	0.335	0.311	0.291		
		taud	2.462	2.421	2.361	2.300	2.305	2.319	2.314	2.356	2.416	2.463	2.472	2.440		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	242	263	272	275	277	277	275	272	268	252	231	223		
		Edh at Noon	19	25	32	37	39	39	38	35	30	24	19	17		
(44) All-Sky Solar Radiation	RadStd		252	475	848	1330	1574	1733	1649	1406	1070	623	294	201		
			39	90	119	161	168	160	180	155	112	53	35	34		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (69 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page